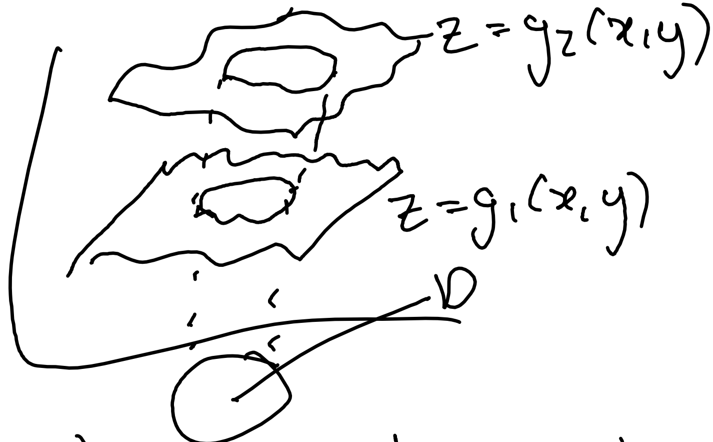
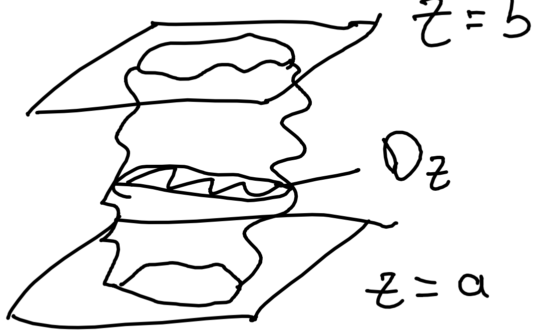
I showed you



$$\iint_D \left(\int_{g_1(x,y)}^{g_2(x,y)} f(x,y,z)\,dz \right) dx\,dy$$

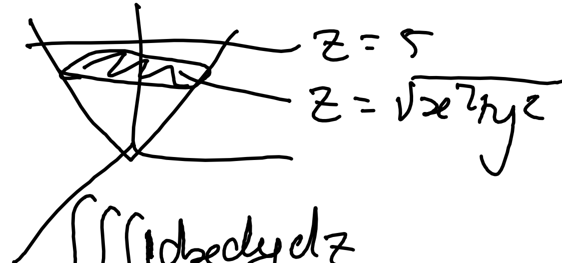
but what about this shape:



as z changes
D_z changes

$$\int_a^b \left(\iint_{D_z} f(x,y,z)\,dx\,dy \right) dz$$

e.g. $z = \sqrt{x^2+y^2}$
cone.



Let's find volume $V(R) = \iiint_R 1\,dxdydz$

we know $V = \pi r^2 \frac{h}{3}$ so we expect $\frac{\pi 5^3}{3}$ ($h = r$ when $z = \sqrt{(x^2+y^2)}$)

$\iint_{D_z} dxdy$? in this case, change of variables

$D_z : z = \sqrt{x^2+y^2} = r$

$0 \le r \le z$

$0 \le \theta \le 2\pi$

$$\int_0^5 \left(\iint_{D_z} 1\, dxdy \right) dz = \int_0^5 \left(\int_0^{2\pi} \int_0^z 1 \cdot r\, dr d\theta \right) dz = \int_0^5 \int_0^{2\pi} \frac{r^2}{2} \Big]_0^z d\theta dz$$

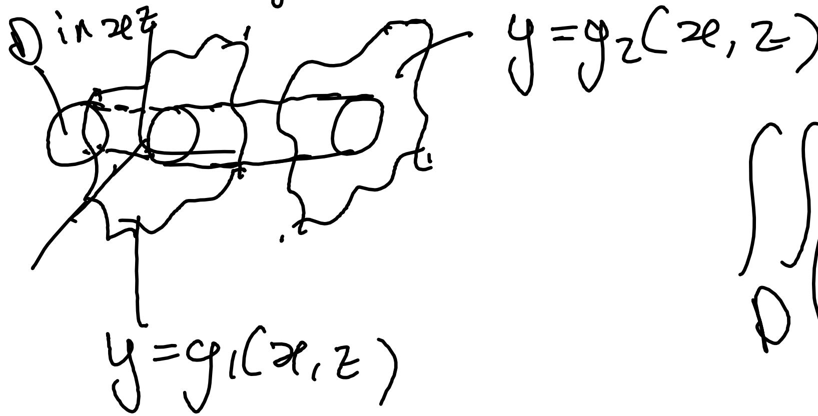
$$= \int_0^5 \int_0^{2\pi} \frac{z^2}{2} d\theta dz = \int_0^5 \frac{z^2}{2} \theta \Big]_0^{2\pi} dz = \int_0^5 z^2 \pi\, dz = \pi \frac{z^3}{3} \Big]_0^5$$

$$= \frac{\pi}{3} 5^3 \; !$$

of course, you could also do

$$\iint_{\text{circle}} \int_{\sqrt{x^2+y^2}}^{5} dz\,dx\,dy$$, and should get the same.

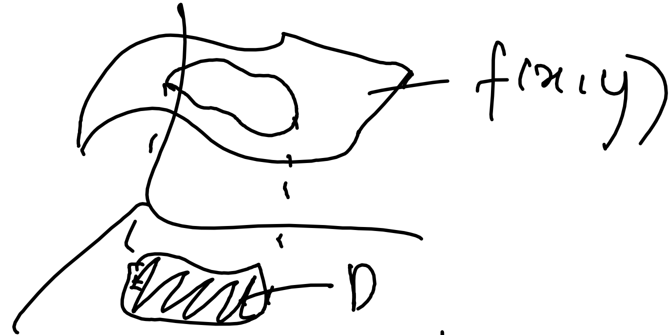
Btw, we've been focusing on z a lot, check Stewart for interest



$$\iint_D \left(\int_{g_1(x,z)}^{g_2(x,z)} f(x,y,z)\,dy \right) dx\,dz$$

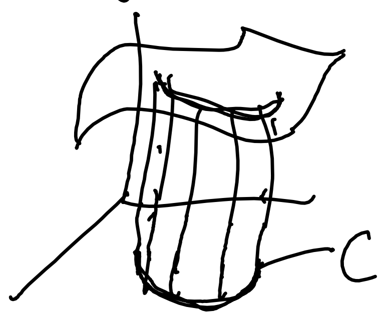
Back to $f(x,y)$!

we've integrated over a <u>domain</u> $\longrightarrow$



which requires infinitesimal areas clearly

What if I just want to add up along a $\boxed{\text{curve}}$:



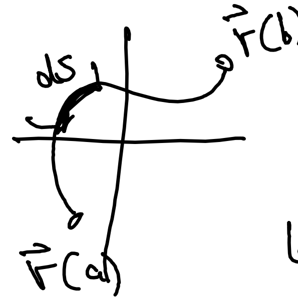
no "inside" points added! we get a new surface,

other way to think of this: $f(x,y)$ assigns a scalar value to every point (x,y), so $f(x,y)$ defines a scalar field.

e.g. f^n gives density per unit length we need the curve C as a vector f^n

$\vec{r} : [a,b] \to \mathbb{R}^2$

$\vec{r}(t) = (r_1(t), r_2(t))$



$s(t)$ is the arclength f^n so ds is infinitesimal length of curve.

recall from Vector-functions-II that

$$\frac{ds}{dt} = \|\vec{r}\,'(t)\| \quad \text{so} \quad ds = \|\vec{r}\,'(t)\|\,dt$$

Thus, to add up the values of $f(x,y)$ as we travel along the curve C, we integrate:

$$\int_C f(x,y)\,ds = \int_a^b f(\vec{r}(t))\,\|\vec{r}\,'(t)\|\,dt$$

formally: the line/ linear integral of a scalar-valued fn: let $f : D \subset \mathbb{R}^n \to \mathbb{R}$ be a continuous fn. Let C be a regular curve described by the vector fn $\vec{r} : [a,b] \to \mathbb{R}^n$, which takes its values in the set D (i.e. $\vec{r}(t) \in D \ \forall t \in [a,b]$). Then, the line integral of the function f along the curve C is defined as

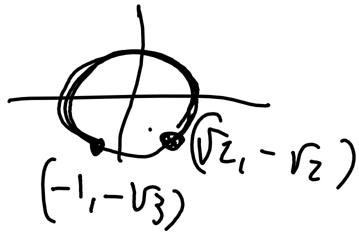
$$\int_C f(x_1, x_2, \dots, x_n)\, ds = \int_a^b f(\vec{r}(t))\, \|\vec{r}\,'(t)\|\, dt$$

example: $\int_C 2x^2\,ds$ $\quad f(x,y) = 2x^2$ where

C is arc of circle w/ rad 2, centered @ origin, (*) from $(\sqrt{2}, -\sqrt{2})$ to $(-1, -\sqrt{3})$ (Δ) traveling counter clockwise

(*) $\Rightarrow$ $\vec{r}(t) = (2\cos t, 2\sin t)$

(Δ) $\Rightarrow$ $a = -\frac{\pi}{4} \qquad b = \frac{4\pi}{3}$



$\vec{r}\,'(t) = (-2\sin t, 2\cos t)$ so $\|\vec{r}\,'(t)\| = \sqrt{4\sin^2 + 4\cos^2}$
$= \sqrt{4} = 2$

$f(\vec{r}(t)) = f(2\cos t, 2\sin t) = 2 \cdot 4\cos^2 t = 8\cos^2 t$

so $\int_C 2x^2\,ds = \int_{-\frac{\pi}{4}}^{4\pi/3} 8\cos^2 t \cdot 2\,dt = \int_{-\frac{\pi}{4}}^{4\pi/3} 8 \cdot 2\cos^2 t\,dt$ ♡

use $\cos 2\theta = 2\cos^2\theta - 1$ so $2\cos^2\theta = \cos 2\theta + 1$

♡ $= \int 8(\cos 2t + 1)\,dt = 8\frac{\sin 2t}{2} + 8t \Big]_{-\pi/4}^{3\pi/4}$

Vector fields now, we let $\vec{F}(x,y)$ assign a vector to each point!

let $n \geq 2$ be an integer. let $D \subset \mathbb{R}^n$. a vector f^n of the type $\vec{F}: D \subset \mathbb{R}^n \to \mathbb{R}^n$ is called a n-dimensional vector field. So $\vec{F}$ assigns to every point $(x_1, x_2, \ldots, x_n) \in D$ a vector w/ n components:

$$\vec{F}(x_1, x_2, \ldots, x_n) = \left(F_1(x_1, x_2, \ldots, x_n), F_2(x_1, x_2, \ldots, x_n), \ldots, F_n(x_1, x_2, \ldots, x_n)\right)$$

we'll work in $n=2$ and $n=3$

in $\mathbb{R}^2$ we often write $\vec{F}(x,y) = (P(x,y), Q(x,y))$

so from a point (x,y) draw the arrow representing $(P(x,y), Q(x,y))$

e.g. $\vec{F}(x,y) = (-y, x)$

famous vector fields: gravitional force between object 1 and object 2, depending on where object 2 is

electric force

<u>gradient vector field</u>: given $f: D \subset \mathbb{R}^2 \to \mathbb{R}$ s.t. $f_x(x,y)$ and $f_y(x,y)$ exist $\forall (x,y) \in D$, $\nabla f(x,y) = (f_x(x,y), f_y(x,y))$ gives a 2D vector field!

sim: $f: D \subset \mathbb{R}^3 \to \mathbb{R}$, $\nabla f(x,y,z) =$

$$(f_x(x,y,z), f_y(x,y,z), f_z(x,y,z))$$